School: _____
Team Number: C - _____
Names: _____

Dynamic Planet Test

Topic: Earth's Fresh Waters

Northmont Invitational

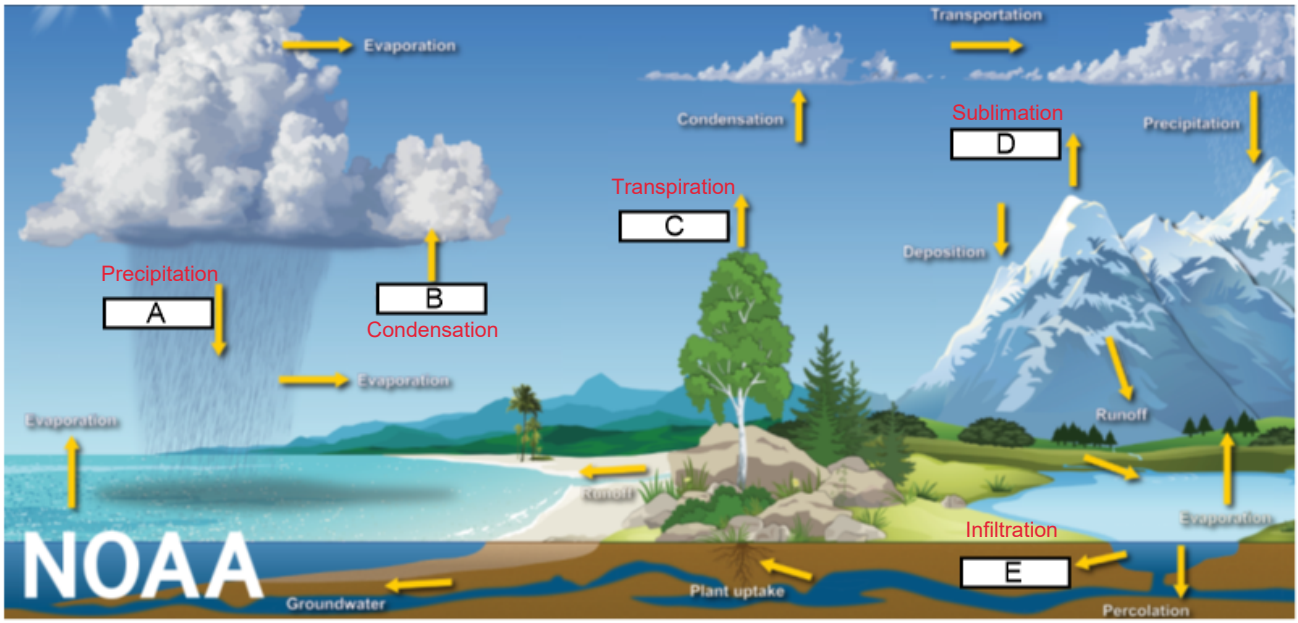
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Division C

Instructions:
You'll have 50 minutes to complete the test.

You are only allowed to use resources prepared beforehand in accordance with the event rules. If a question is worth multiple points, it will be shown with the question, otherwise assume 1 point per answer. Ties will be broken by a predetermined set of questions. Unanswered questions are counted as zeros. You won't lose points for wrong questions. Written responses should be answered as clearly and as fully as possible or you risk lost points. Good Luck!

1. Name the missing steps of the water cycle. (1 point each)



2. What are the percentages of earth's water, per NOAA resources? (1 point each)
   a. Ocean 96.5 %
   b. Fresh Water 2.5 %

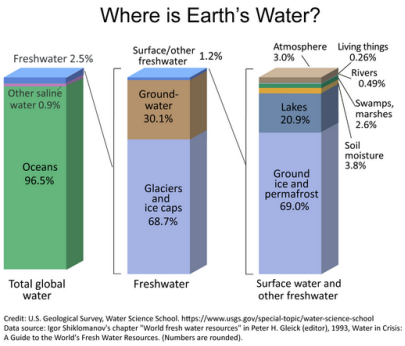
See diagram below.

3. Of Earth's freshwater, what are the percentages of each, per NOAA resources? (1 point each)
   a. Surface Water 1.2 %
   b. Ground Water 30.1 %
   c. Glaciers 68.7 %

See diagram below.

4. Of the surface water, what are the percentages of each per NOAA resources? (1pt each)
   a. Rivers 0.49 %
   b. Lakes 20.9 %
   c. Ground Ice/Permafrost 69.0 %
   d. Atmosphere 3.0 %
   e. Other 6.61 %

See diagram to the right.



5. The acronym NOAA stands for:
   a. National Ocean and Atmospheric Association
   b. National Observance and Analysis of the Atmosphere
   c. Nearly Objective Atmosphere Analysis
   d. Noticing Objects and Aliens in the Atmosphere

6. The main source of the hydrologic cycle is:
    a. Lakes
    b. Glaciers
    c. Ocean
    d. Rivers

7. True or False: About 20% of the world's precipitation falls on land.

8. Water that moves over the earth's surface is called:
    a. Tributary
    b. Runoff
    c. Gradient
    d. Rills

9. The rate at which surface materials can absorb water is called:
    a. Runoff capacity
    b. Transpiration capacity
    c. Percolation capacity
    d. Infiltration capacity

10. Which of these is the largest channel flow?
    a. Rill
    b. River
    c. Creek
    d. Stream

11. The headwaters of a river is 1000 m above sea level. The horizontal distance of the river is 500 km. What is the average gradient of the river system?
    a. 0.5 m/km
    b. 0.5 m
    c. 2 m
    d. 2 m/km

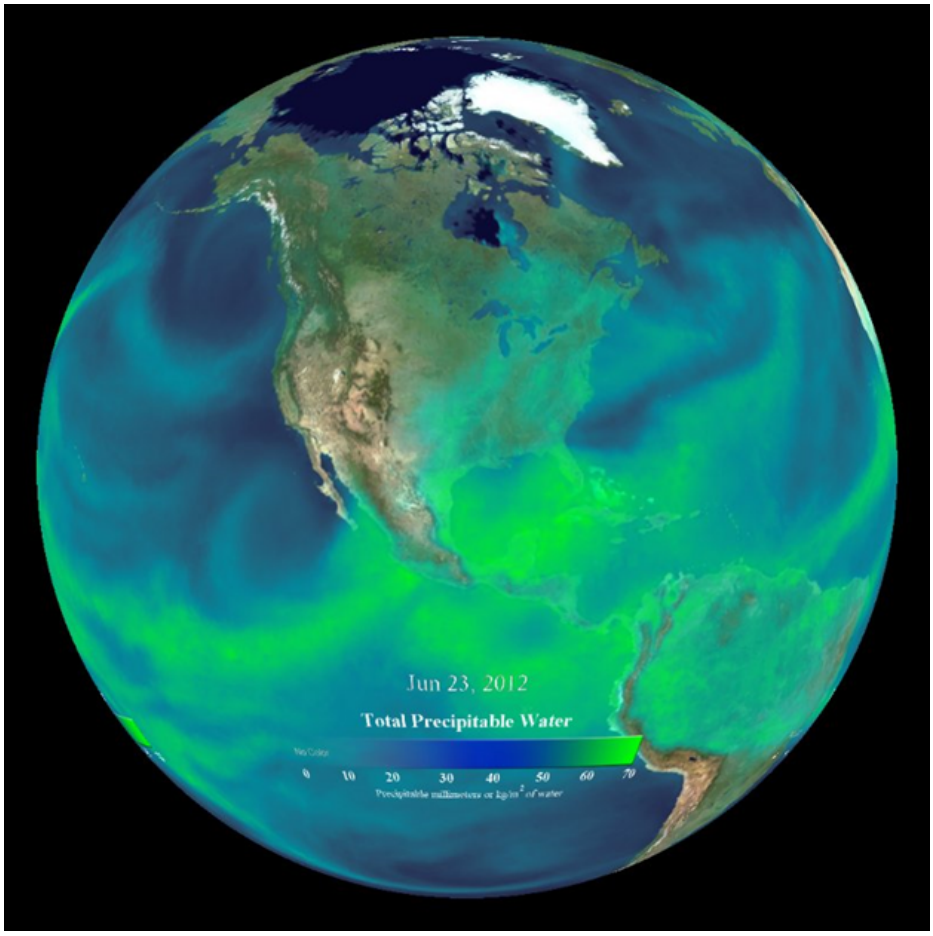
Gradient = Slope (m)
m= change in vertical / change in horizontal
change in vertical = 1000m
change in horizontal = 500 km
m=1000m / 500 km = 2 m/km

12. The volume of water passing a particular point in a given period of time is called:
    a. gradient
    b. velocity
    c. discharge
    d. turbulence

discharge = area x velocity
Discharge is the volume of water moving down a stream or river per unit of time, commonly expressed in cubic feet per second or gallons per day.

13. True or False: For most river systems, the river's average velocity is faster downstream than upstream.

Figure 2: Please use this image for the questions that refer to it.



Credit: NOAA

14. The NOAA can create images and animations like the one shown in Figure 2. What technology helps to create such images?
    a. Radar
    b. Ground laser sensors
    c. Satellites
    d. Phone app technology

15. In Figure 2, the _____ is being sensed to create this image?
    a. Microwave radiation
    b. Infrared radiation
    c. Moisture
    d. Pressure

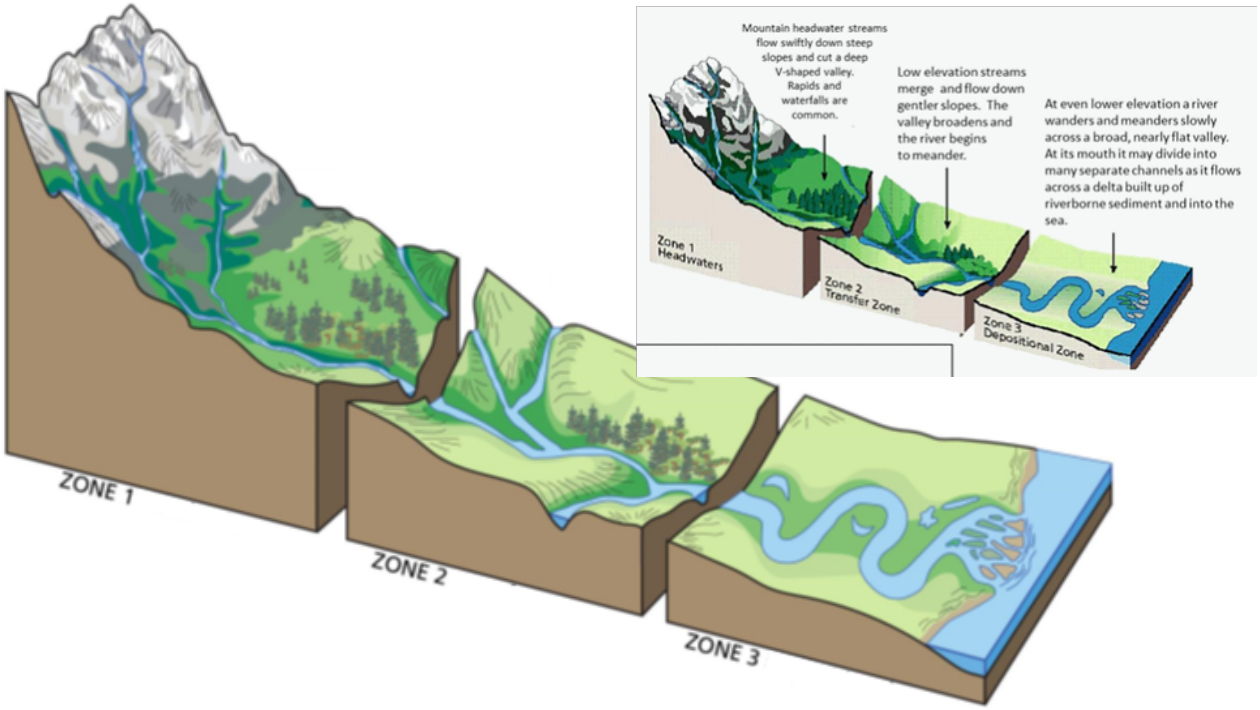
16. True or False: Figure 2 shows light precipitable water southwest of Florida.

    The darker green means more precipitation.
    Note the color guide at the bottom of the picture (Total Precipitable Water)

17. True or False: Figure 2 shows very little precipitable water over California.

    The blue over California means little to no precipitation.
    Note the color guide at the bottom of the picture (Total Precipitable Water)

Figure 3: Use this image for the questions that refer to it.



18. Use Figure 3. Which zone would be known as the head waters? Zone 1

19. Use Figure 3. Which zone would have the most deposition? Zone 3

20. Use Figure 3. Which zone shows meandering? Zone 3

21. Use Figure 3. Which zone would have the most V-shaped riverbed? Zone 1

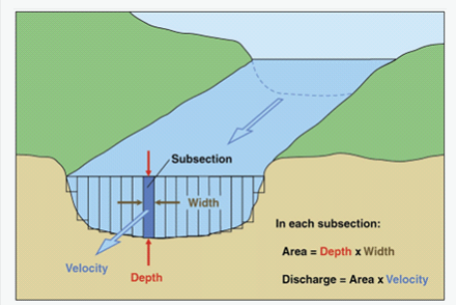
22. Use Figure 3. In which zone do the valleys begin to broaden? Zone 2

23. A stream has a width of 20 feet and a depth of 5 feet. Its cross-sectional area is:
    a. 15 ft
    b. 25 ft^2
    c. 100 ft^2
    d. 4 ft^2

Area = width * depth
Width = 20ft
Depth = 5 ft
(Cross-sectional) Area = 20ft * 5ft = 100 ft^2

24. A stream has a cross-sectional area of 100 ft^2 and the velocity of this stream is 10 feet per second. The discharge of the stream is:
    a. 10 ft^2 per second
    b. 10 ft per second
    c. 10,000 ft^2 per second
    d. 1,000 ft^3 per second

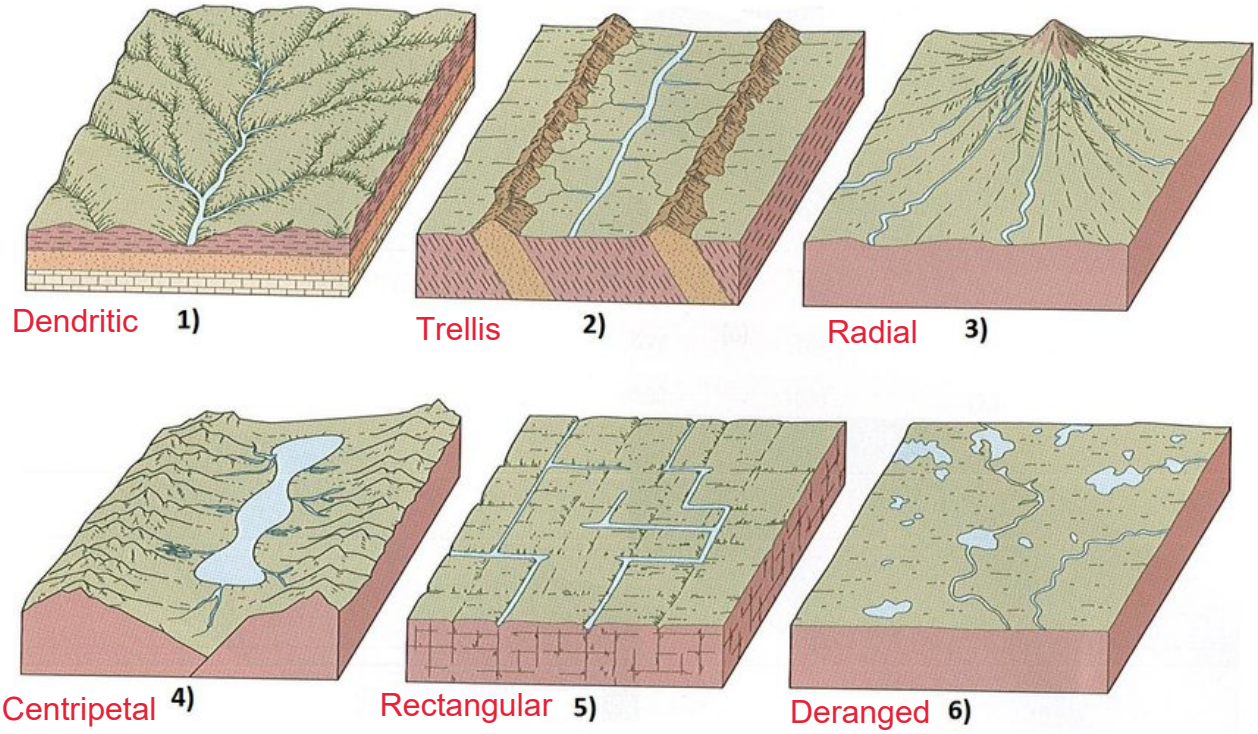
Discharge = Area * Velocity
Area = 100 ft^2
Velocity = 10 ft/sec
Discharge=100 ft^2 * 10ft
Discharge = 1,000 ft^3/sec



25. A stream has a discharge of 25,000 cubic feet per second. The width of the stream is 100 feet and the depth is 5 feet. What is the velocity of the stream? 4 points: showing work is required for full points.

Q=V*A or V=Q/A
A=100 ft x 5 ft = 500 ft^2
V=250000 ft^3/s / 500 ft^2
V=50 ft/s

Figure 4: Use this image for the questions that refer to it.



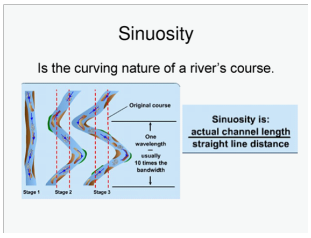
26. Which image in Figure 4 represents a radial drainage pattern? 3

27. Which image in Figure 4 represents a dendritic drainage pattern? 1

28. In Figure 4, what do you call the drainage pattern of #5? Rectangular drainage

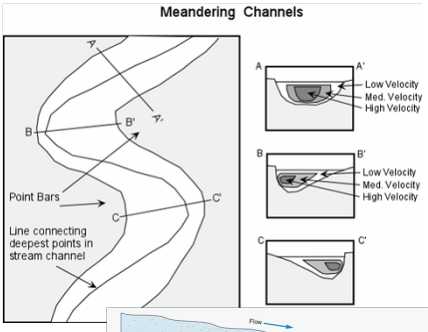
29. In Figure 4, what do you call the drainage pattern of #6? Deranged drainage

30. What is sinuosity?
    a. the area of water in a channel cross section multiplied by the average velocity of water in that cross section.
    b. the length along the stream divided by the straight-line distance along the stream valley.
    c. The stream's vertical drop divided by the horizontal distance of the stream.
    d. the volume of water moving down a stream or river per unit of time



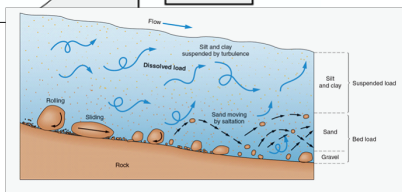
31. Along straight stretches of a stream channel, the highest velocities of water are found in what part of the stream?

a. near the water surface in the center of the stream
b. near the water surface along the stream banks
c. along the stream bottom in the center of the stream
d. along the stream bottom near the stream banks



32. The suspended load of a stream ____.

a. is deposited before the bed load
b. consists primarily of highly soluble substances
c. moves along the bottom of the channel by rolling or sliding
d. usually consists of fine sand, silt, and clay-sized particles



33. The ability of a stream to erode and transport material depends largely on its ____.

a. width
b. length
c. velocity
d. density

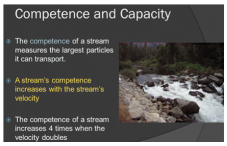
Carrying Capacity

- The ability of a stream to transport material is called the carrying capacity of the stream.
- Carrying capacity is dependent of stream velocity and discharge.
- Discharge is the volume of water that passes a specific point in a given period of time. We can calculate discharge as:

Discharge = width x depth x velocity
(m^3/s) (m) (m) (m/s)

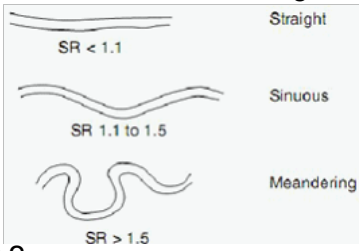
34. What is the measure of the largest particles a stream can carry?

a. competence
b. capacity
c. discharge
d. gradient



35. If the sinuosity is _____, the river is considered to be meandering.

a. 0 - 0.5
b. 0.5 - 1.0
c. 1.5 or greater
d. 5.0 or greater

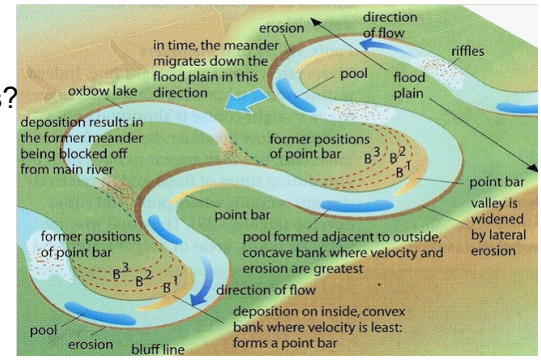


36. For a meandering river, which statement is true?

a. The speed is greatest on the outside of the curve, causing erosion.
b. The speed is greatest on the inside of the curve, causing deposition..
c. The speed is slowest on the outside of the curve causing deposition
d. The speed is slowest on the inside of the curve, causing erosion.

37. What are the two main types or freshwater wetlands?

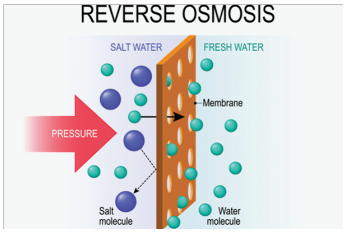
a. lakes and ponds
b. rivers and streams
c. littoral zones and benthic zones
d. marshes and swamps



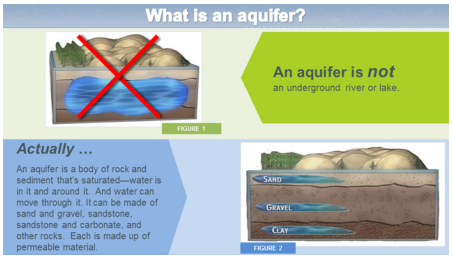
38. Which of the following statements is not correct?
    a. Lakes may contain a region that receives little sunlight.
    b. Wetlands help remove Pollutants from water.
    c. Swamps are dominated by woody shrubs and water-loving trees.
    d. Rivers generally move faster, and their oxygen levels decrease, as they near the ocean.

39. Swamps are commonly found on
    a. rocky hillsides.
    b. flat, poorly drained land.
    c. forest clearings.
    d. bottoms of ponds or lakes

40. One method of desalination uses high pressure to force saltwater through a membrane filter. This method is called:
    a. diffusion
    b. distillation
    c. reverse osmosis
    d. active transport

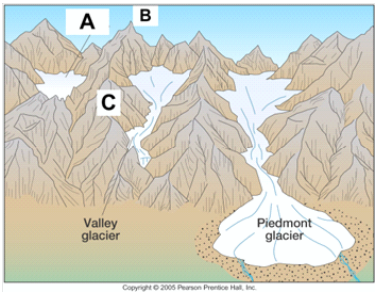


41. The geological layer, consisting of underground caverns and porous layers of sand, gravel, or bedrock, where groundwater flows, is called:
    a. the zone of saturation
    b. the water table
    c. an aquifer
    d. surface water



42. Karst landscapes occur in areas high in
    a. Calcium
    b. Limestone
    c. shale
    d. sandstone

43. Which of the following is a characteristic of valley glaciers?
    a. They move downslope.
    b. They form over broad regions.
    c. They are larger than continental glaciers.
    d. They completely melt each summer.



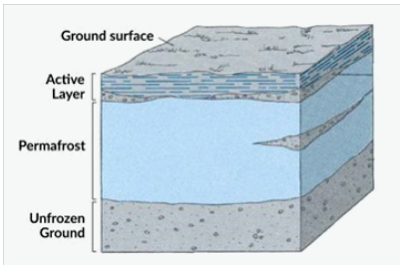
44. Most pollutants that enter estuaries break down over time, yet they are still considered a problem because
    a. the pollutants eventually concentrate In the ocean.
    b. it is the volume of the pollutants, more than the time factor, that harms estuaries.
    c. the pollutants break down into still more dangerous toxins.
    d. None of the above

45. Which of the following is a point source of pollution?

a. Fertilizer from a farming area
b. Runoff from city streets
c. A wastewater pipe leak
d. Leaking septic tanks

46. The maximum seasonal thaw depth in permafrost is called the

a. active layer
b. permeable layer
c. variable layer
d. Interchange layer



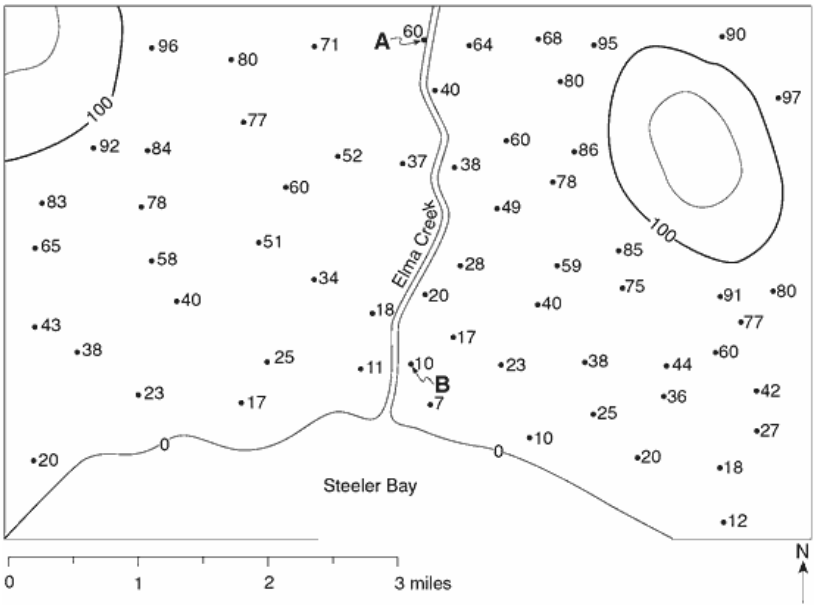
47. What is a major result of urban development?

a. Less runoff
b. More impervious surface
c. Flooding
d. More runoff

48. What does USGS stand for?

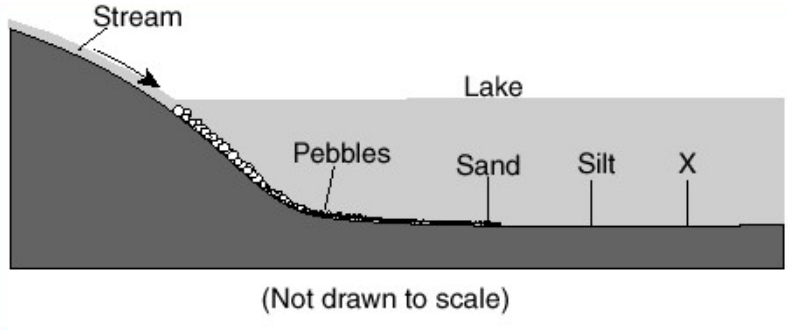
a. Under Saturated Ground Systems
b. Undeveloped Sinuosity Gradient Systems
c. United States Geological Services
d. United States Geological Survey

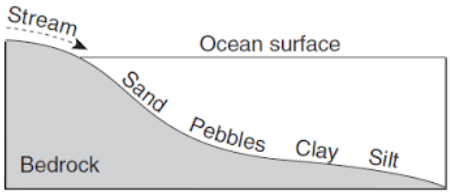
49. Using the map below, report the direction that Elma Creek flows. South



The numbers are elevations.
Water flows down gradient (downhill).
Note the Steeler Bay is at elevation 0.

50. Name the type of sediment for point X. Clay





51. Examine the image below. Name this water feature.



Braided stream/river/channel

52. Examine the image below. Name this feature.



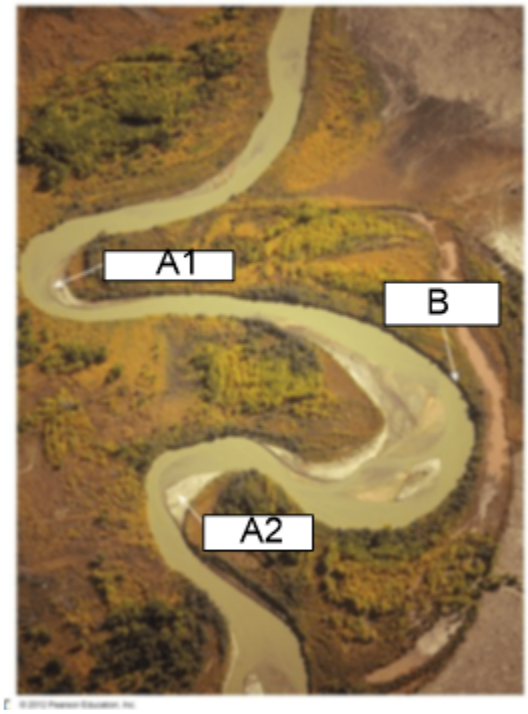
Delta

53. Examine the image below. Name this water feature.



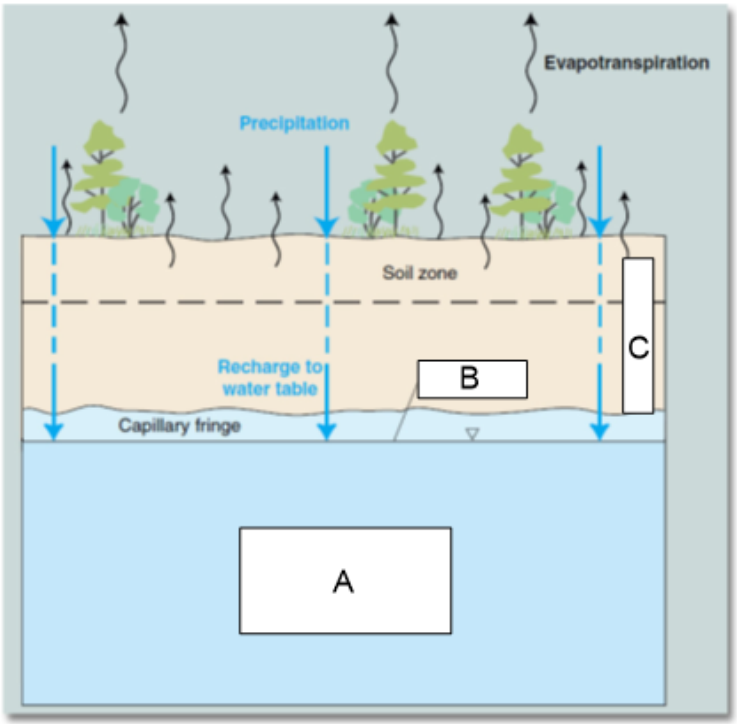
Meandering stream/river/channel

54. Examine the image below. Name the specific features A & B. Then choose the one location that has the most potential to become an oxbow lake.

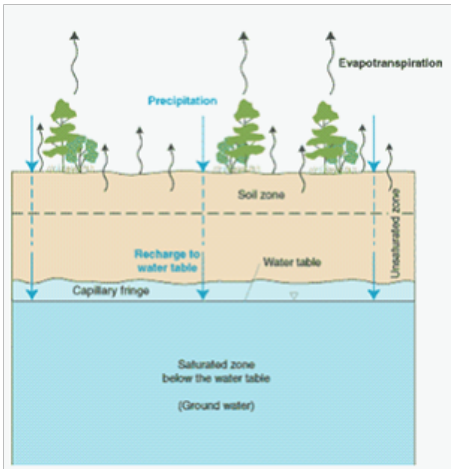


A - Point Bar
B - Cut Bank
A1 - most likely to become an oxbow lake

55. Name A, B and C.



A - Groundwater / Saturated Zone
B - Water Table
C - Unsaturated Zone

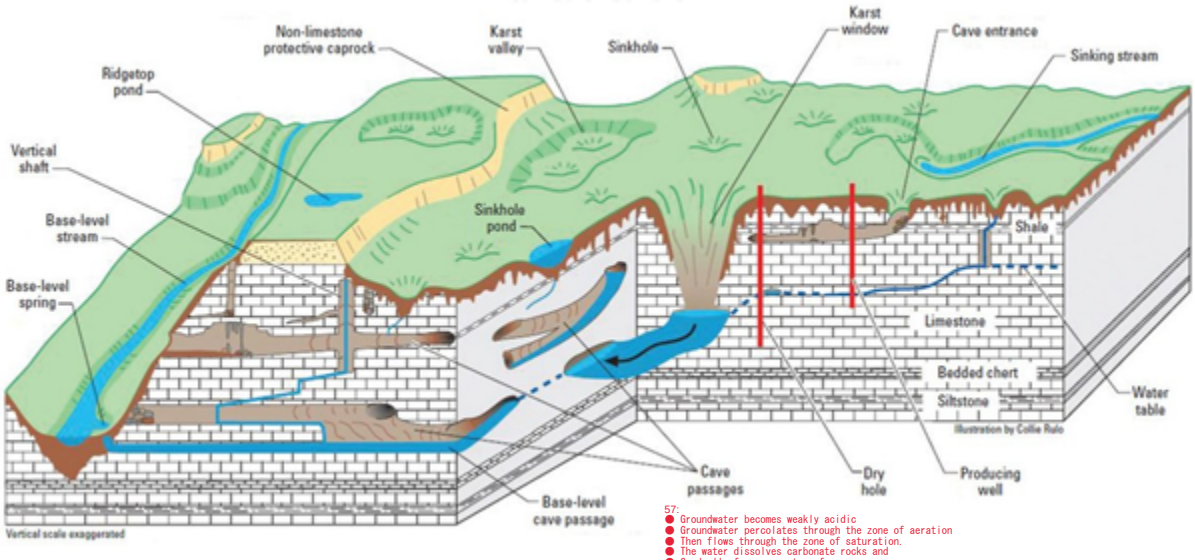


WRITTEN RESPONSES

56. Complete the data table that compares the various types of lakes, their origins and their characteristics. 8 points.

Types of Lake	Where it forms	Shape
A. Oxbow	A1. Flood plain area near river	A2. Crescent shape
B. Cirque	B1. Tops of mountains	B2. Bowl shape
C. Kettle	C1. Fields / flat ground	C2. Deep depression
D. Glacial	D1. Previously glaciated land	D2. Groove or depression left by glacier

57. Use the following diagram to explain how caves form.



57:
- Groundwater becomes weakly acidic
- Groundwater percolates through the zone of aeration
- Then flows through the zone of saturation.
- The water dissolves carbonate rocks and
- Gradually forms a system of passageways.
- The water moves along the water table and
- The dissolved water is carried to the surface stream.
- When the water table drops, it leaves the abandoned channel ways which we know as caves or caverns.

1

58. Explain how Glaciers change the shape of valleys. (8 points) Type text here

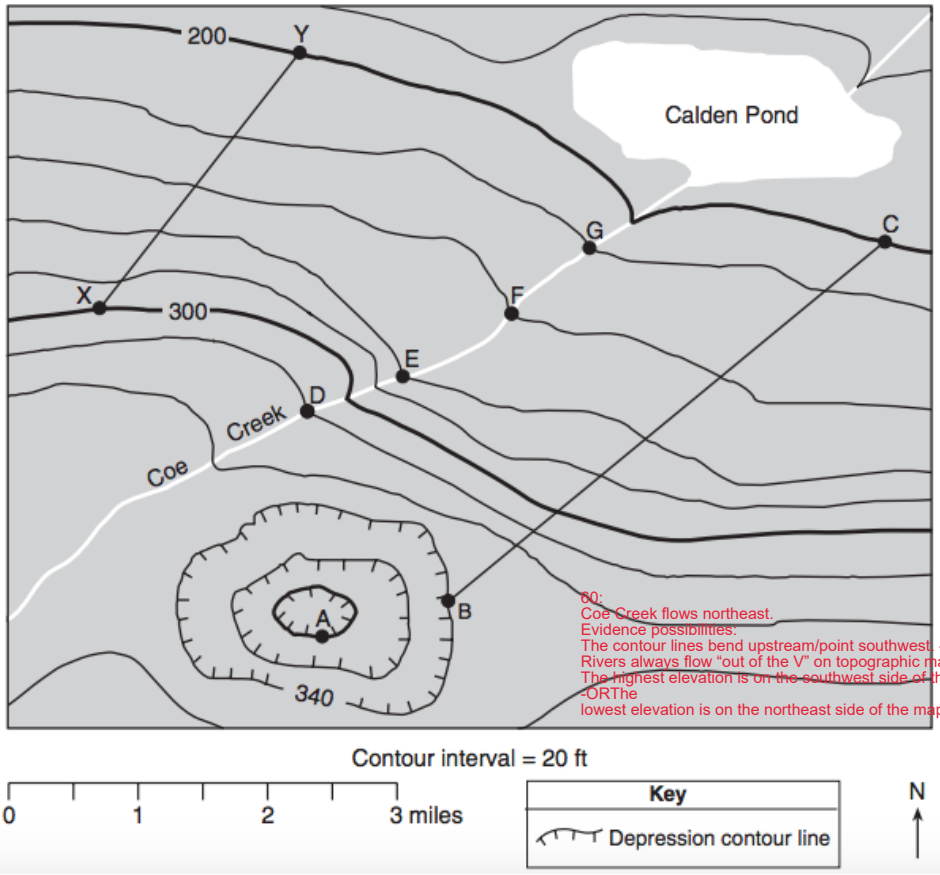
- A glacier flows over land and picks up rocks
- -Called plucking
- Weight of ice breaks the rocks
- Rock fragments freeze to bottom of glacier
- Glacier drags rocks across land/valley.
- This gouges and scratches the bedrock
- -called abrasion
- Over time, the glacier creates a U-shaped valley.

59. Explain the formation of alluvial fans. (10 points)

59:
- Form best on lowlands with adjacent highlands
- Arid and semi-arid regions
- Little vegetation
- Periodic rainstorms
- Surface materials are quickly saturated and
- runoff is funneled into a mountain canyon.
- The runoff cannot spread laterally here, but
- when it discharges onto the lowlands, it quickly spreads out.
- This is a repeated process.
- The "fan" is actually lobate deposits of alluvium on land

1

The topographic map below shows elevations in feet. Point A represents a location on Earth's surface. Lines BC and XY are reference lines on the map. Points D, E, F, and G represent locations along Coe Creek.



1. Note the elevations. They are higher in the bottom left (SW) and lower in the top right (NE). Water flows downgradient (downhill).
2. In contour lines, the 'V's point upstream. In this case, they point SW, so the water flows NE.

60:
Coe Creek flows northeast.
Evidence possibilities:
The contour lines bend upstream/point southwest - OR -
Rivers always flow "out of the V" on topographic maps.
The highest elevation is on the southwest side of the map and rivers flow downhill.
-ORThe
lowest elevation is on the northeast side of the map and rivers flow downhill.

60. State the direction that Coe Creek flows and give 2 pieces of evidence that support your answer. (3 points) See answer above.

61. Where will the water flow fastest along Coe Creek: between points D and E or between points F and G? Support your answer with evidence from the map. (3 points)

See answer below.

62. Where on the map has the potential to hold freshwater at the surface? Why? (2 points)

See answer below.

61:
The Water will flow the fastest between points D and E.
Evidence possibilities:
There are more contour lines between D and E
More contour lines means it is steeper between D and E
It is less steep between F and G.
There are less contour lines or (less of a difference in elevation) between F & G.

62:
Point A has potential to hold fresh water because it is in the deepest part of a depression with no drainage pattern shown on the surface.